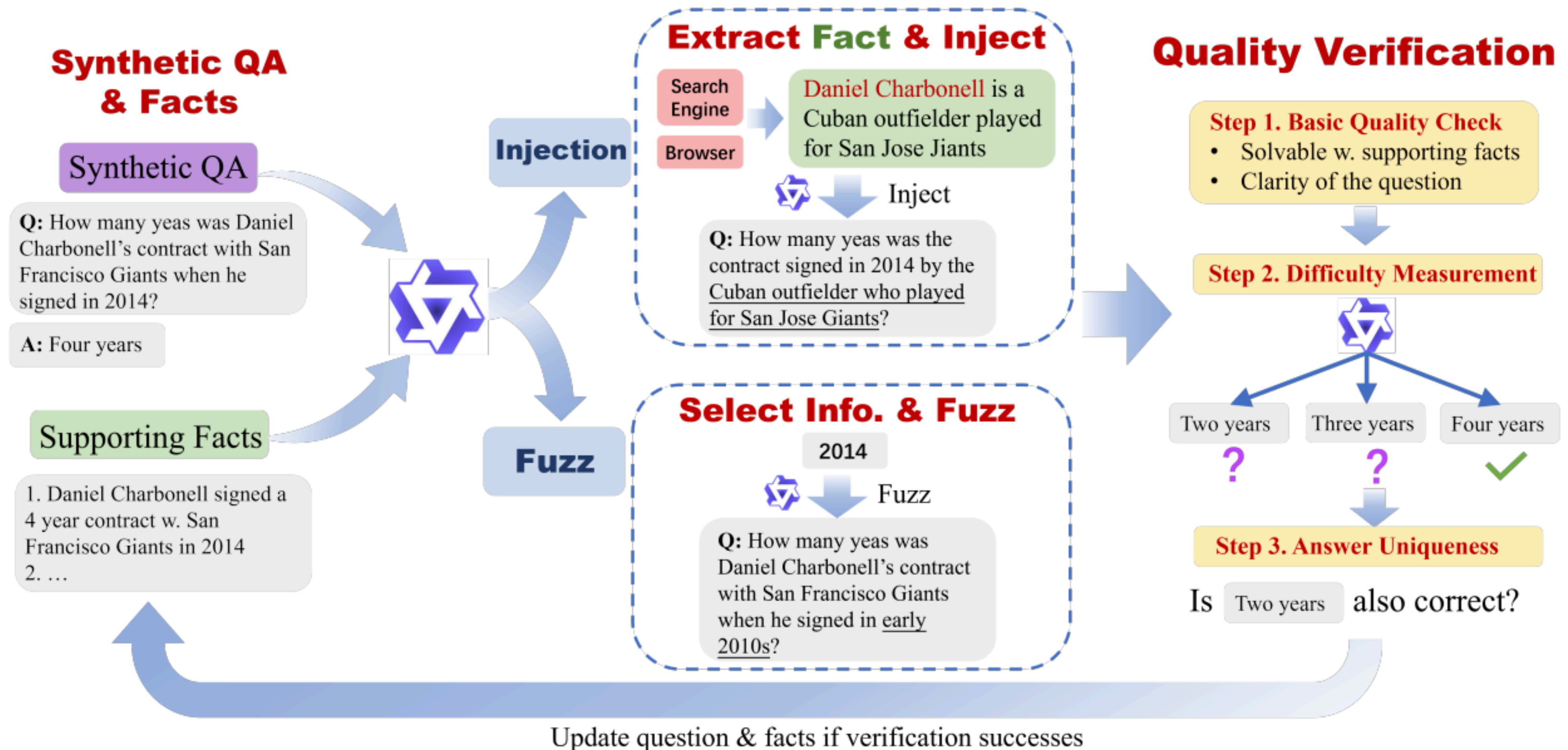
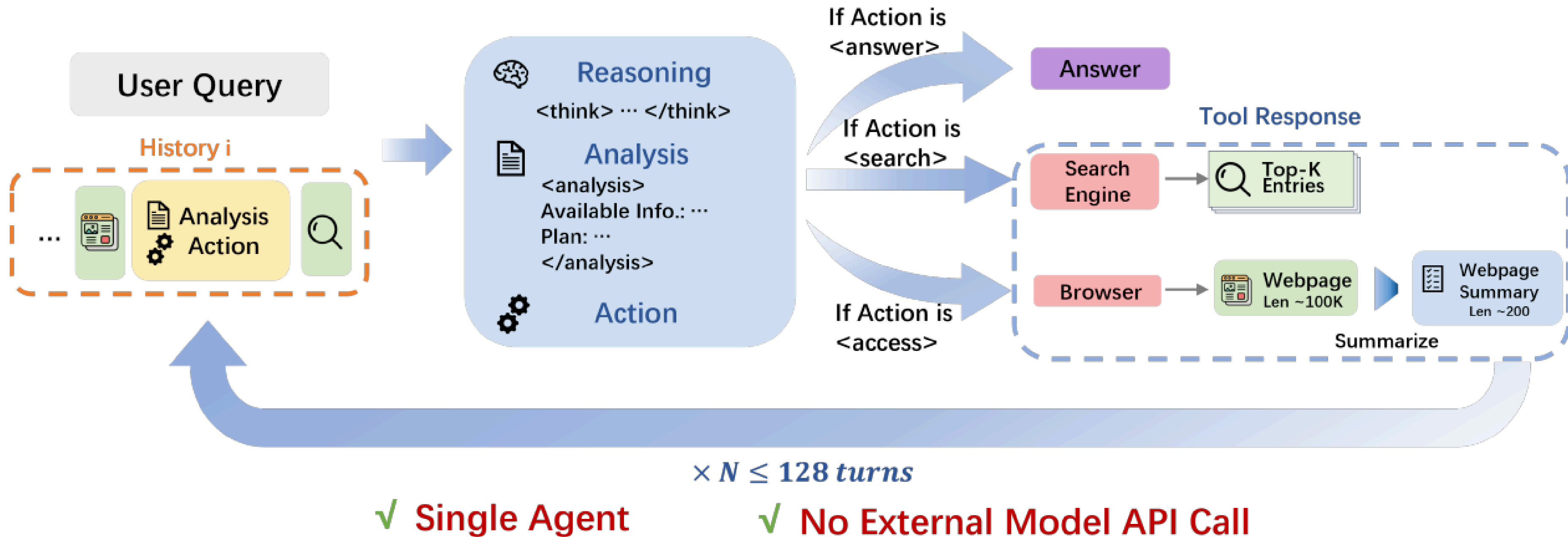


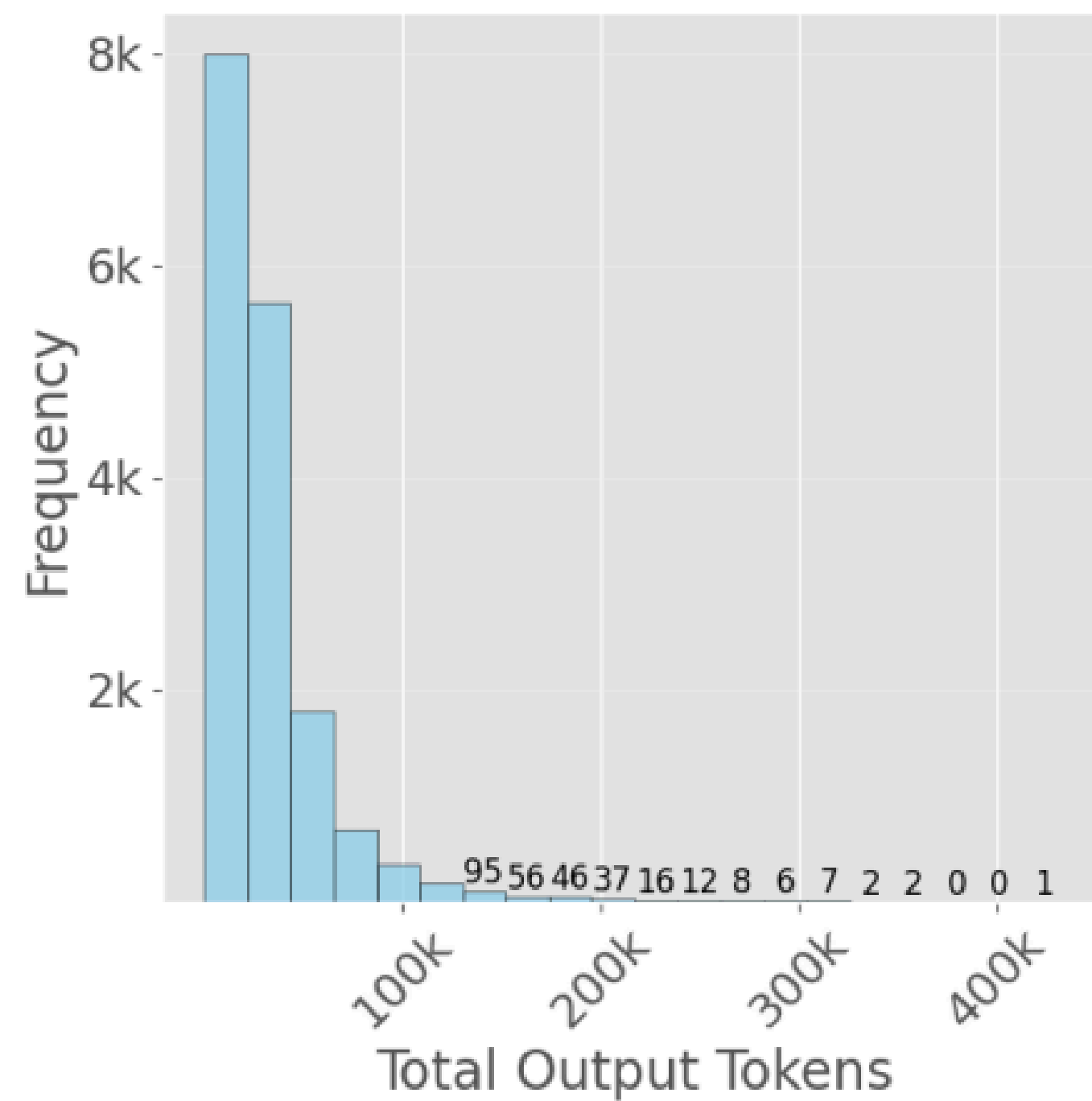
Jak generowane są trudniejsze pytania?



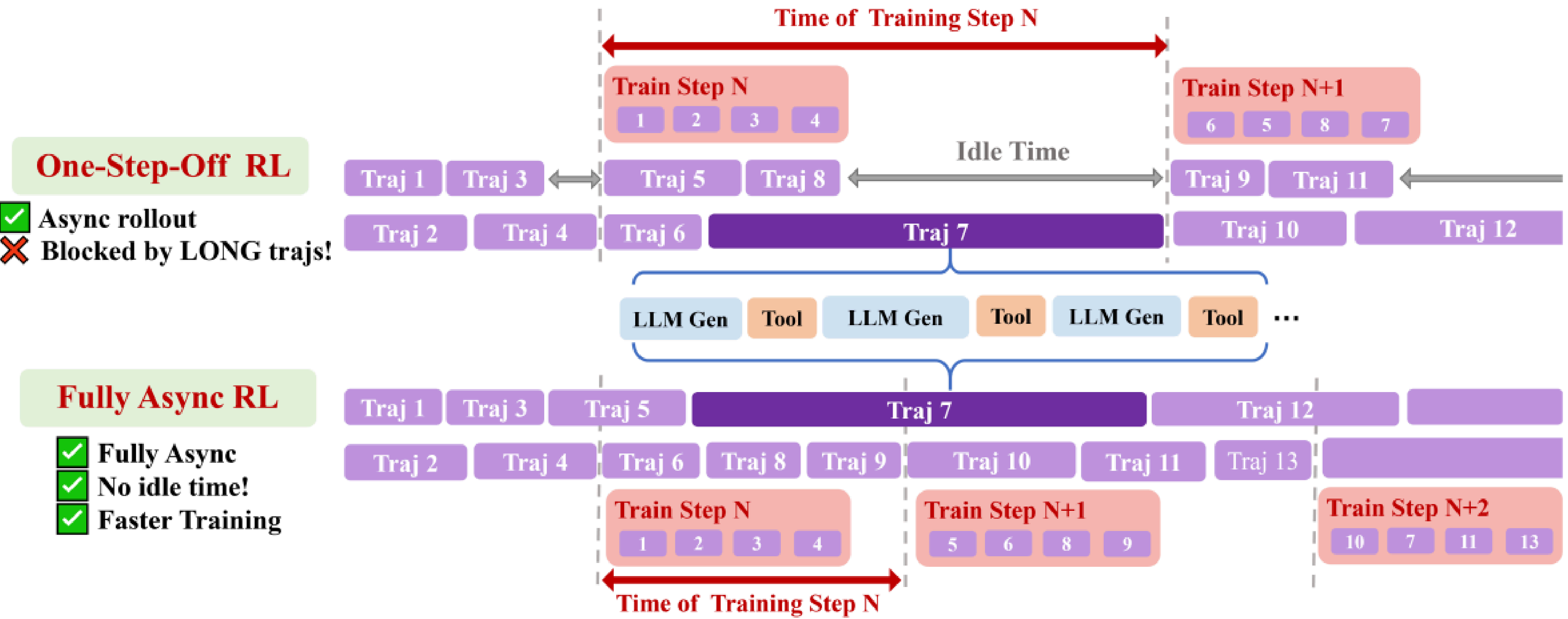
Pętla generowania odpowiedzi



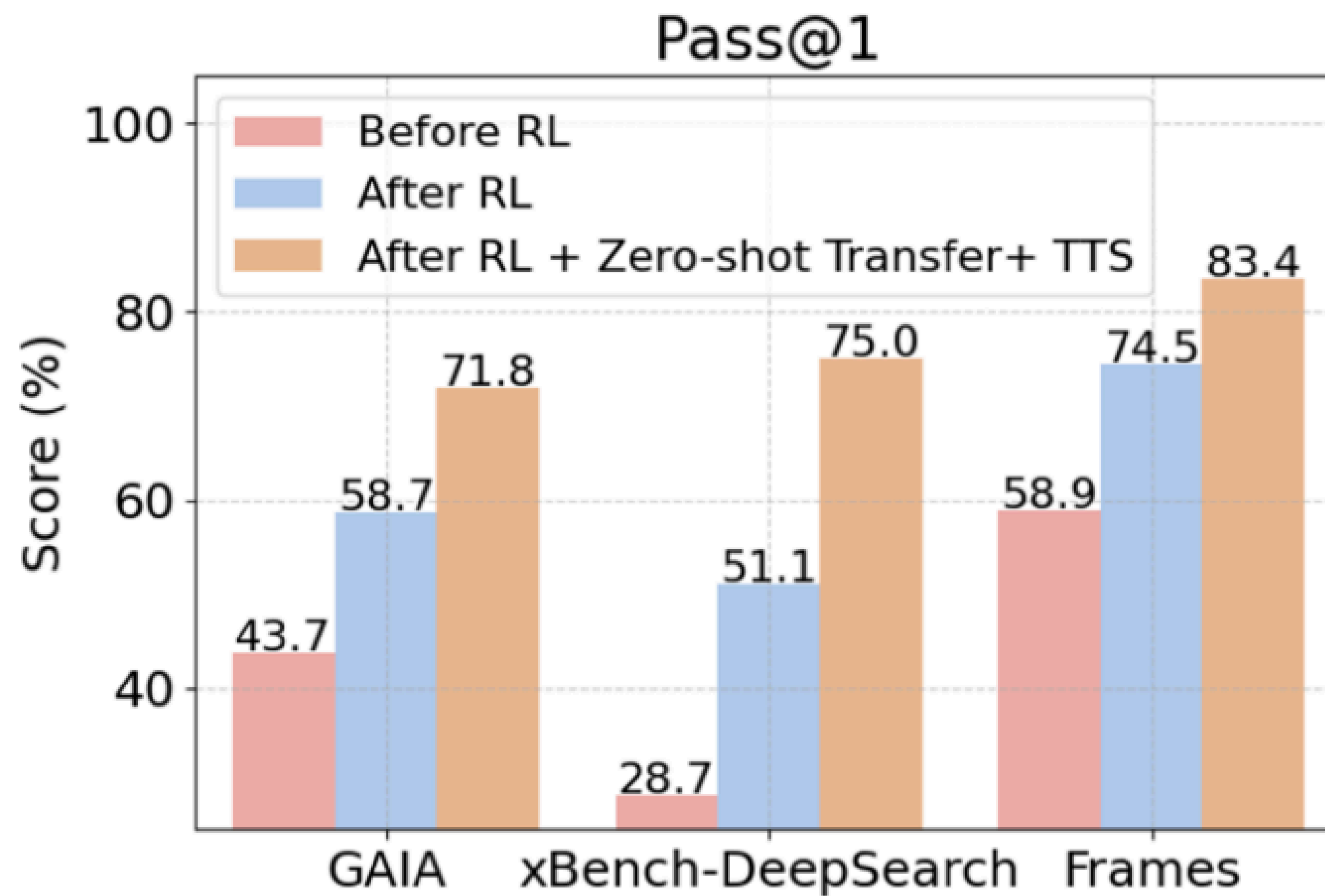
Dystrybucja długości outputów (widać outliery, które mają po paraset tysięcy tokenów)



Uczenie asynchroniczne



Wyniki na różnych benchmarkach



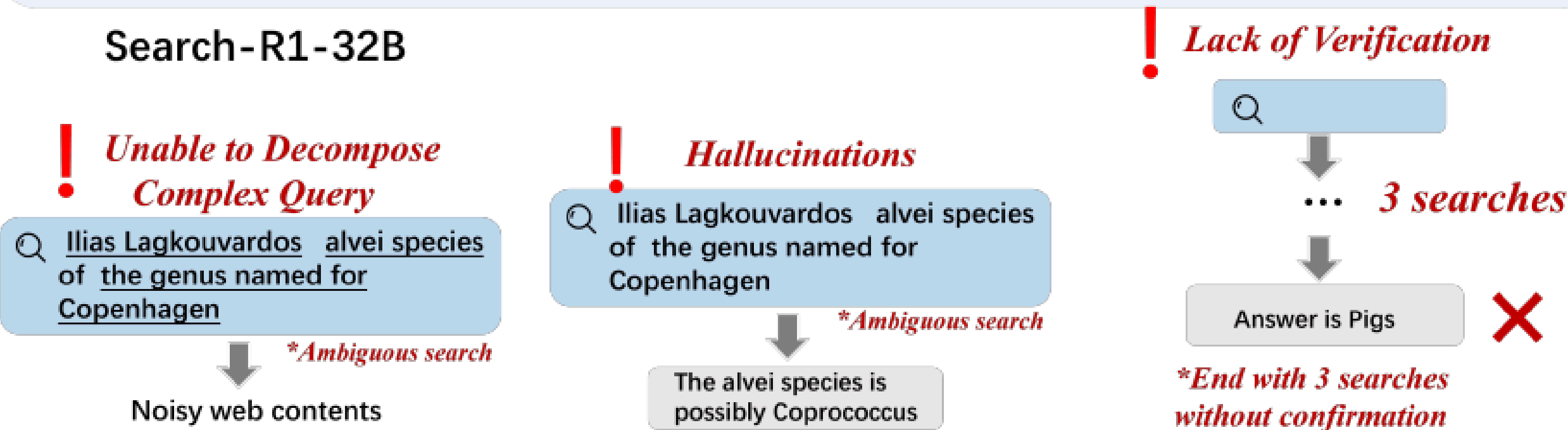
Method	No Commercial LLM	Non-Search Tools	GAIA	xBench-DeepSearch	Frames	HLE
Commercial Deep Research Agents						
Kimi-Researcher	-	-	-	69.0 [†]	78.8 [†]	26.9 [†]
OpenAI DeepResearch	-	-	67.0 [†]	-	-	26.6 [†]
General LLMs using Tools						
OpenAI-o3	-	-	70.5 [†]	66.7 [†]	84.0 [†]	20.2 [†]
Qwen3-30B-A3B	-	-	35.9 [†]	32.0 [†]	56.4 [†]	13.2 [†]
Qwen3-235B-A22B	-	-	45.6 [†]	46.0 [†]	-	20.0 [†]
DeepSeek-R1	-	-	-	55.0 [†]	82.0 [†]	24.8 [†]
Claude-4-Sonnet	-	-	68.3 [†]	64.6 [†]	80.7 [†]	20.3 [†]
Open-source Search Agents						
Search-o1 (QwQ)	✓	-	48.1	40.3	63.6	-
Search-R1-32B	✓	-	28.6	19.5	44.1	-
WebThinker-QwQ	✓	-	48.5 [†]	32.8	57.7	15.8 [†]
Simple DS-QwQ	✓	-	50.5 [†]	35.8	68.8 [†]	-
WebDancer-QwQ	✗	-	51.5 [†]	40.0	63.8	-
WebSailor-32B	✓	-	53.2 [†]	53.3[†]	69.8 [†]	-
WebShaper-32B	✓	-	53.3 [†]	-	-	-
AFM-RL-32B	✗	Code Sandbox	55.3 [†]	-	-	18.0 [†]
MiroThinker-32B-DPO	✗	Visual Question Answering, Audio Transcription, Linux Sandbox	60.9 [†]	56.0 [†]	74.8 [†]	20.6 [†]
Ours (QwQ-32B)						
<i>ASearcher</i>	✓	-	58.7	51.1	74.5	21.5
+ Summary=DeepSeek-V3	✗	-	60.3	56.4	76.6	23.4
+ Test-time Search (K=16)	✗	-	71.8	75.0	83.4	24.6

An Example Question from GAIA

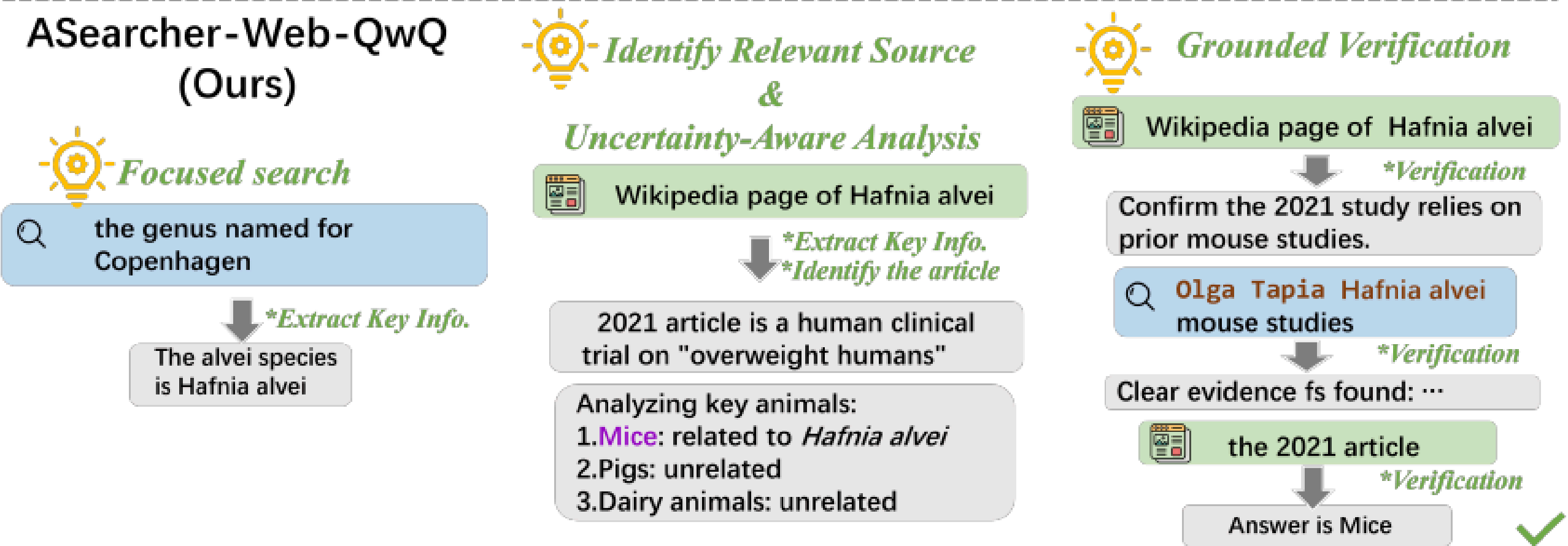
Q: What animals that were mentioned in :
• both Ilias Lagkouvardos's and Olga Tapia's papers on the alvei species of the genus named for Copenhagen outside the bibliographies
• and also in the 2021 article cited on the alvei species' Wikipedia page about a multicenter, randomized, double-blind study?

A: Mice

Search-R1-32B



ASearcher-Web-QwQ (Ours)



Zjawiska wyłaniające się (podczas fazy reasoning)

4.3 EMERGENT BEHAVIORS

